© 2021 ASHRAE, Inc.

ALMATY, Kazakhstan

WMO: **368700**

Lat: **43.2394N** Lon: **76.9328E** Elev: **851** StdP: **91.51** Time Zone: **6.00 (E06)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB 99.6%	Heating DB 99%	Humidification DP/MCDB and HR 99.6% DP	99.6% HR	99.6% MCDB	99% DP	99% HR	99% MCDB	Coldest Month WS/MCDB 0.4% WS	0.4% MCDB	1% WS	1% MCDB	MCWS/PCWD to 99.6% DB MCWS	PCWD	WSF	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	**1**	**-19.8**	**-16.8**	**-21.2**	**0.6**	**-19.1**	**-18.8**	**0.8**	**-16.5**	**6.4**	**-1.2**	**5.3**	**-2.2**	**1.1**	**170**	**0.458**	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB 0.4% DB	0.4% MCWB	1% DB	1% MCWB	2% DB	2% MCWB	Evaporation WB/MCDB 0.4% WB	0.4% MCDB	1% WB	1% MCDB	2% WB	2% MCDB	MCWS/PCWD to 0.4% DB MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	**7**	**12.9**	**34.4**	**18.5**	**32.9**	**18.1**	**31.2**	**17.8**	**20.5**	**30.1**	**19.6**	**29.3**	**18.8**	**28.7**	**2.2**	**20**	(2)

	Dehumidification DP/MCDB and HR 0.4% DP	0.4% HR	0.4% MCDB	1% DP	1% HR	1% MCDB	2% DP	2% HR	2% MCDB	Enthalpy/MCDB 0.4% Enth	0.4% MCDB	1% Enth	1% MCDB	2% Enth	2% MCDB	Extreme Max WB	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	**17.2**	**13.6**	**24.3**	**16.2**	**12.7**	**23.7**	**15.2**	**11.9**	**23.0**	**63.2**	**30.4**	**59.8**	**29.3**	**57.1**	**29.0**	**27.3**	(3)

Extreme Annual Design Conditions

	Extreme Annual WS 1%	2.5%	5%		Extreme Annual Temperature Mean Min	Mean Max	Standard Deviation Min	Standard Deviation Max	n-Year Return Period Values of Extreme Temperature n=5 years Min	n=5 years Max	n=10 years Min	n=10 years Max	n=20 years Min	n=20 years Max	n=50 years Min	n=50 years Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	**6.2**	**5.0**	**4.2**	DB	**-22.3**	**37.7**	**4.3**	**1.9**	**-25.4**	**39.0**	**-27.9**	**40.1**	**-30.3**	**41.2**	**-33.4**	**42.6**	(4)
(5)				WB	**-22.4**	**22.5**	**4.3**	**1.8**	**-25.5**	**23.8**	**-28.1**	**24.9**	**-30.5**	**26.0**	**-33.6**	**27.3**	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	**Temperatures, Degree-Days and Degree-Hours**	DBAvg		**10.2**	**-5.3**	**-3.1**	**4.8**	**11.7**	**17.0**	**21.7**	**24.0**	**23.1**	**17.8**	**10.6**	**3.0**	**-3.2**	(6)
(7)		DBStd		**11.30**	**5.51**	**5.57**	**5.87**	**4.80**	**3.78**	**3.20**	**2.95**	**3.37**	**4.32**	**4.73**	**5.38**	**5.16**	(7)
(8)		HDD10.0		**1733**	**474**	**367**	**177**	**37**	**2**	**0**	**0**	**0**	**2**	**50**	**216**	**410**	(8)
(9)		HDD18.3		**3472**	**732**	**599**	**419**	**202**	**71**	**8**	**1**	**4**	**62**	**243**	**461**	**668**	(9)
(10)		CDD10.0		**1821**	**0**	**0**	**16**	**88**	**217**	**352**	**435**	**406**	**234**	**68**	**5**	**0**	(10)
(11)		CDD18.3		**518**	**0**	**0**	**0**	**3**	**28**	**110**	**178**	**151**	**45**	**3**	**0**	**0**	(11)
(12)		CDH23.3		**5749**	**0**	**0**	**2**	**59**	**291**	**1154**	**1993**	**1694**	**514**	**42**	**1**	**0**	(12)
(13)		CDH26.7		**2272**	**0**	**0**	**0**	**10**	**64**	**416**	**874**	**737**	**164**	**6**	**0**	**0**	(13)
(14)	**Wind**	WSAvg		**1.6**	**1.3**	**1.4**	**1.7**	**2.0**	**2.0**	**1.9**	**1.9**	**1.8**	**1.6**	**1.4**	**1.4**	**1.3**	(14)
(15)	**Precipitation**	PrecAvg		**649**	**34**	**38**	**71**	**104**	**105**	**61**	**38**	**29**	**28**	**57**	**55**	**38**	(15)
(16)		PrecMax		**1013**	**70**	**68**	**154**	**222**	**214**	**193**	**128**	**74**	**98**	**148**	**126**	**83**	(16)
(17)		PrecMin		**359**	**8**	**5**	**27**	**1**	**24**	**8**	**2**	**0**	**0**	**4**	**8**	**6**	(17)
(18)		PrecStd		**123**	**15**	**15**	**32**	**53**	**46**	**34**	**28**	**20**	**21**	**36**	**26**	**20**	(18)
(19)	**Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures**	0.4%	DB		**8.3**	**12.1**	**21.9**	**28.0**	**31.0**	**34.2**	**36.9**	**36.1**	**32.9**	**27.2**	**20.0**	**9.9**	(19)
(20)			MCWB		**3.4**	**6.5**	**11.4**	**14.7**	**17.4**	**18.2**	**19.9**	**18.9**	**17.0**	**14.0**	**10.4**	**4.9**	(20)
(21)		2%	DB		**5.5**	**8.7**	**19.1**	**24.9**	**28.2**	**32.2**	**34.8**	**34.2**	**30.4**	**24.0**	**16.3**	**7.2**	(21)
(22)			MCWB		**2.0**	**4.5**	**10.1**	**14.1**	**16.9**	**18.3**	**18.7**	**18.4**	**16.2**	**12.9**	**9.3**	**3.5**	(22)
(23)		5%	DB		**3.5**	**6.2**	**16.4**	**22.5**	**26.5**	**30.8**	**33.0**	**32.6**	**28.5**	**21.2**	**13.1**	**5.1**	(23)
(24)			MCWB		**1.0**	**3.0**	**9.2**	**12.9**	**16.3**	**18.0**	**18.4**	**17.9**	**15.4**	**11.7**	**7.9**	**2.2**	(24)
(25)		10%	DB		**1.8**	**4.0**	**13.8**	**20.1**	**24.8**	**29.1**	**31.2**	**30.9**	**26.4**	**18.8**	**10.7**	**3.2**	(25)
(26)			MCWB		**0.1**	**1.6**	**8.2**	**11.9**	**15.5**	**17.5**	**18.1**	**17.4**	**14.6**	**10.8**	**6.7**	**1.2**	(26)
(27)	**Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures**	0.4%	WB		**4.2**	**6.8**	**12.6**	**17.0**	**19.2**	**21.2**	**22.2**	**21.7**	**19.0**	**14.7**	**12.0**	**6.1**	(27)
(28)			MCDB		**6.6**	**10.4**	**19.4**	**25.0**	**27.4**	**29.9**	**31.6**	**31.2**	**30.8**	**25.1**	**16.9**	**8.6**	(28)
(29)		2%	WB		**2.6**	**4.8**	**11.0**	**15.2**	**18.0**	**19.9**	**20.6**	**19.9**	**17.1**	**13.4**	**9.7**	**3.9**	(29)
(30)			MCDB		**4.9**	**8.2**	**17.7**	**22.1**	**26.1**	**28.8**	**30.6**	**30.3**	**28.1**	**21.9**	**14.8**	**6.3**	(30)
(31)		5%	WB		**1.3**	**3.2**	**9.6**	**13.7**	**17.0**	**18.9**	**19.7**	**18.9**	**16.0**	**12.5**	**8.3**	**2.5**	(31)
(32)			MCDB		**3.0**	**5.8**	**15.4**	**20.7**	**25.0**	**28.3**	**29.5**	**29.8**	**26.6**	**19.7**	**12.4**	**4.7**	(32)
(33)		10%	WB		**0.1**	**1.8**	**8.4**	**12.6**	**16.1**	**18.2**	**18.9**	**18.1**	**15.0**	**11.4**	**6.9**	**1.2**	(33)
(34)			MCDB		**1.7**	**3.6**	**13.4**	**18.9**	**23.4**	**27.2**	**28.9**	**28.7**	**25.3**	**17.8**	**10.2**	**3.0**	(34)
(35)	**Mean Daily Temperature Range**		MDBR		**9.7**	**9.7**	**10.4**	**11.5**	**11.9**	**12.5**	**12.9**	**13.6**	**13.4**	**12.3**	**9.6**	**9.0**	(35)
(36)		5% DB	MCDBR		**11.6**	**12.1**	**14.8**	**15.7**	**14.7**	**14.9**	**15.3**	**15.7**	**15.9**	**16.2**	**14.4**	**11.1**	(36)
(37)			MCWBR		**8.4**	**8.5**	**8.1**	**7.9**	**7.3**	**6.5**	**6.1**	**6.2**	**6.8**	**7.7**	**8.6**	**7.9**	(37)
(38)		5% WB	MCDBR		**10.4**	**11.0**	**13.3**	**13.9**	**13.5**	**13.3**	**13.7**	**14.9**	**14.7**	**14.2**	**13.0**	**10.3**	(38)
(39)			MCWBR		**7.7**	**7.8**	**7.6**	**7.5**	**7.2**	**6.6**	**6.4**	**6.5**	**6.8**	**7.2**	**8.1**	**7.5**	(39)
(40)	**Clear-Sky Solar Irradiance**	taub			**0.318**	**0.362**	**0.422**	**0.459**	**0.435**	**0.437**	**0.431**	**0.404**	**0.388**	**0.391**	**0.357**	**0.312**	(40)
(41)		taud			**2.163**	**2.055**	**1.969**	**1.955**	**2.103**	**2.158**	**2.201**	**2.268**	**2.253**	**2.132**	**2.139**	**2.188**	(41)
(42)		Ebn at Noon			**787**	**807**	**802**	**806**	**840**	**838**	**839**	**851**	**834**	**766**	**727**	**754**	(42)
(43)		Edh at Noon			**97**	**129**	**160**	**176**	**156**	**148**	**140**	**127**	**119**	**118**	**99**	**86**	(43)
(44)	**All-Sky Solar Radiation**	RadAvg			**2.02**	**2.91**	**3.82**	**4.93**	**6.16**	**6.54**	**6.48**	**6.05**	**4.86**	**3.19**	**1.97**	**1.57**	(44)
(45)		RadStd			**0.15**	**0.24**	**0.29**	**0.42**	**0.41**	**0.27**	**0.38**	**0.30**	**0.26**	**0.27**	**0.19**	**0.13**	(45)

Historical Trends

		DBAvg	Heating 99% DB	Heating 99% DP	Cooling 1% DB	Cooling 1% WB	Cooling 1% DP	Degree-Days HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	**+0.91**	**+2.33**	**N/A**	**N/A**	**-0.88**	**-1.40**	**-203**	**-252**	**+139**	**+72**	(46)
(47)	Regional (7 neighbors)	**N/A**	**N/A**	**N/A**	**N/A**	**-0.52**	**-0.89**	**N/A**	**N/A**	**N/A**	**N/A**	(47)

Nomenclature: See separate page